

```
aws cloudwatch put-metric-data --namespace "MyApp" --metric-name "ActiveUsers" --value 120
```

custom metrics Practicals

Launch and ec2 instance

```
>>sudo apt install amazon-cloudwatch-agent.x86_64
```

```
Last login: Sat Nov  1 13:15:56 2025 from 18.206.107.28
[ec2-user@ip-172-31-16-129 ~]$ sudo yum install amazon-cloudwatch-agent.x86_64
Last metadata expiration check: 1:01:55 ago on Sat Nov  1 13:16:35 2025.
Dependencies resolved.
=====
Package                                Architecture          Version
=====
Installing:
amazon-cloudwatch-agent                x86_64                1.300059.1-1.am
Transaction Summary
=====
```

Will create a script for custom metrics

```
>>vi custom_metrics.sh
```

```
#!/bin/bash
```

```
# Set the necessary variables
```

```
INSTANCE_ID="i-04280bae119f24fd2"
```

```
NAMESPACE="demo_Metrics"
```

```
# Get memory utilization percentage
```

```
MEMORY_UTILIZATION=$(free | grep Mem | awk '{printf "%.2f\n", $3/$2 * 100}')
```

```
# Publish custom metric to CloudWatch
```

```
aws cloudwatch put-metric-data \
```

```
--metric-name "MemoryUtilization" \
```

```
--namespace "$NAMESPACE" \
```

```
--dimensions "InstanceId=$INSTANCE_ID" \
```

```
--value "$MEMORY_UTILIZATION" \
```

```
--unit "Percent"
```

```
#!/bin/bash

# Set the necessary variables
INSTANCE_ID="i-04280bae119f24fd2"
NAMESPACE="Roshan_Metrics"

# Get memory utilization percentage
MEMORY_UTILIZATION=$(free | grep Mem | awk '{printf "%.2f\n", $3/$2 * 100}')

# Publish custom metric to CloudWatch
aws cloudwatch put-metric-data \
  --metric-name "MemoryUtilization" \
  --namespace "$NAMESPACE" \
  --dimensions "InstanceId=$INSTANCE_ID" \
  --value "$MEMORY_UTILIZATION" \
  --unit "Percent"
```

chmod +x custom_metrics.sh

[ec2-user@ip-172-31-16-129 ~]\$./custom_metrics.sh

If you run the script and get the error then attach the iam role

```
[ec2-user@ip-172-31-16-129 ~]$ chmod +x custom_metrics.sh
[ec2-user@ip-172-31-16-129 ~]$ ./custom_metrics.sh

Unable to locate credentials. You can configure credentials by running "aws configure".
[ec2-user@ip-172-31-16-129 ~]$
```

Attach iam role to your instance

```
[ec2-user@ip-172-31-16-129 ~]$ ./custom_metrics.sh
[ec2-user@ip-172-31-16-129 ~]$
```

Role dhakkan created.
View role

Creation date November 01, 2025, 19:59 (UTC+05:30)	ARN arn:aws:iam::457650192798:role/dhakkan	Instance profile ARN arn:aws:iam::457650192798:instance-profile/dhakkan
Last activity -	Maximum session duration 1 hour	

Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions

Permissions policies (2)
Info

You can attach up to 10 managed policies.

Search

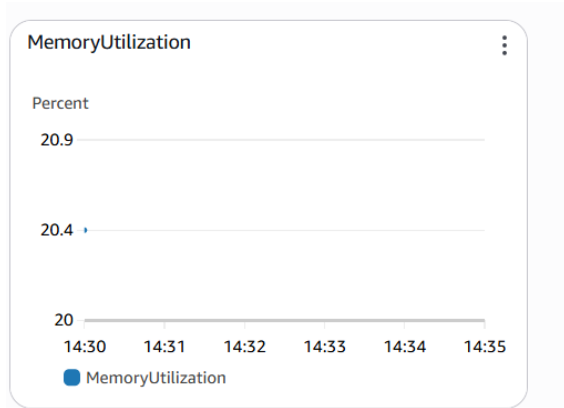
Filter by Type
All types

☐

☐

Policy name	Type	Attached entities
CloudWatchAgentAdminPolicy	AWS managed	1
CloudWatchAgentServerPolicy	AWS managed	1

In your cloud watch you can see your metrias



This way you can create you custom metrics

One more metrics

```
aws cloudwatch put-metric-data \  
  --namespace "demo" \  
  --metric-name "COut" \  
  --unit Count \  
  --value 1 \  
  --dimensions APIName="UserCreate"
```

```
aws cloudwatch put-metric-data \  
  --namespace "ServerMonitoring" \  
  --metric-name "CustomCPUUsage" \  
  --unit Percent \  
  --value 75
```

```
aws cloudwatch put-metric-data \  
  --namespace "ServerMonitoring" \  
  --metric-name "MemoryUsage" \  
  --unit Percent \  
  --value 68
```

For dynamic

```
MEMORY=$(free | awk '/Mem/ {printf("%.2f"), $3/$2 * 100}')
```

```
aws cloudwatch put-metric-data \  
  --namespace "ServerMonitoring" \  
  --metric-name "MemoryUsage" \  
  --unit Percent \  
  --value $MEMORY
```

```
CPU=$(top -bn1 | grep "Cpu(s)" | awk '{print $2 + $4}')
```

```
aws cloudwatch put-metric-data \  
  --namespace "ServerMonitoring" \  
  --metric-name "CPUUsage" \  
  --unit Percent \  
  --value $CPU
```

```
DISK=$(df / | awk 'NR==2 {print $5}' | sed 's/%//')
```

```
aws cloudwatch put-metric-data \  
  --namespace "ServerMonitoring" \  
  --metric-name "DiskUsage" \  
  --unit Percent \  
  --value $DISK
```

Now try to add user

```
Disk Requirements:  
  At least 3633MB more space needed on the / filesystem.  
  
[ec2-user@ip-172-31-16-129 ~]$ aws cloudwatch put-metric-data \  
  --namespace "Roshan" \  
  --metric-name "Count" \  
  --unit Count \  
  --value 1 \  
  --dimensions APIName="UserCreat"  
[ec2-user@ip-172-31-16-129 ~]$  
[ec2-user@ip-172-31-16-129 ~]$ useradd mantu  
useradd: Permission denied.  
useradd: cannot lock /etc/passwd; try again later.  
[ec2-user@ip-172-31-16-129 ~]$ sudo useradd mantu  
[ec2-user@ip-172-31-16-129 ~]$ sudo useradd chantu  
[ec2-user@ip-172-31-16-129 ~]$
```

Now check your metrics

COUNT



1



COUNT